

Repeated-Measures Factorial Analysis

Method, Scenario, and Method × Scenario effects

This supplement presents the corrected paper's primary inference. The ten RunID/seed indices form complete matched blocks, and Method, Scenario, and their interaction are evaluated jointly.

Study	9 methods × 4 scenarios × 10 paired repetitions
Navigation data	360 valid runs; 360 successful; 0 excluded; 0 failed
Source	Corrected manuscript supplied for this update (13 September 2026)

Agri-Social Navigation in Vineyard Corridors

1. Model and design structure

For each continuous endpoint, m indexes method, s indexes scenario, and r indexes the RunID/seed block. The same seed appears in all 36 Method × Scenario cells, so matched dependence is preserved instead of pooling scenarios as independent observations.

$$Y_{msr} = \beta_0 + \beta_m^{(M)} + \beta_s^{(S)} + \beta_{ms}^{(MS)} + b_r + \varepsilon_{msr}$$

Displayed degrees of freedom are uncorrected. Reported p values use the Greenhouse-Geisser (GG) correction where applicable. Partial η^2 quantifies omnibus effect magnitude.

2. Factorial results

Metric	Effect	F	df	GG p	partial η^2
T	Method	887.17	8,72	< 10 ⁻²⁹	0.990
T	Scenario	39.70	3,27	1.19 × 10 ⁻⁸	0.815
T	Method × Scenario	8.21	24,216	5.05 × 10 ⁻⁵	0.477
L	Method	3822.70	8,72	< 10 ⁻⁴⁰	0.998
L	Scenario	6.93	3,27	0.0045	0.435
L	Method × Scenario	2.10	24,216	0.0647	0.190
dmin	Method	55.37	8,72	1.38 × 10 ⁻⁷	0.860
dmin	Scenario	113.42	3,27	1.19 × 10 ⁻¹²	0.926
dmin	Method × Scenario	16.17	24,216	2.76 × 10 ⁻⁸	0.642
TP	Method	146.50	8,72	< 10 ⁻²⁰	0.942
TP	Scenario	1000.23	3,27	< 10 ⁻¹⁵	0.991
TP	Method × Scenario	57.31	24,216	< 10 ⁻¹⁵	0.864
σv	Method	1357.90	8,72	< 10 ⁻²⁴	0.993
σv	Scenario	14.80	3,27	1.10 × 10 ⁻⁵	0.622
σv	Method × Scenario	8.06	24,216	9.96 × 10 ⁻⁵	0.472

3. Endpoint interpretation

- Mission time T: Method, Scenario, and interaction are significant; a controller's time advantage depends on the scenario.
- Path length L: Method and Scenario are significant, but the interaction is not robust under GG (p = 0.0647).
- Minimum distance dmin: the interaction is strong (partial η^2 = 0.642), confirming that separation patterns change across E1-E4.
- Personal-zone exposure TP: this endpoint has the largest interaction (partial η^2 = 0.864).
- Velocity variability σv: Method, Scenario, and interaction are significant; it should not be reduced to one pooled comparison.

4. Mixed-effects sensitivity

As a sensitivity check, a linear mixed-effects model was fitted with Method, Scenario, and Method × Scenario as fixed effects and RunID/seed as a random intercept. Its conclusions agree with the repeated-measures analysis for T, dmin, TP, and σv.

For L, the mixed model yields a significant Wald interaction (p = 8.59 × 10⁻⁴), whereas the GG-corrected repeated-measures test yields p = 0.0647. The paper adopts the conservative conclusion: the path-length interaction is not robustly established.

5. Why pooled omnibus testing is insufficient

A nine-group Welch ANOVA on pooled means would ignore repeated seed structure and could confound method effects with scenario differences. The current inference uses all 360 run-level observations, preserves pairing, and models the interaction explicitly.

6. Reproducibility control

- Preserve all ten RunID/seed blocks and the complete 36-cell design.
- Compute GG p values without replacing displayed uncorrected degrees of freedom with rounded alternatives.
- Report partial η^2 together with F and p.
- Do not select only the significant result when GG and the mixed model disagree.